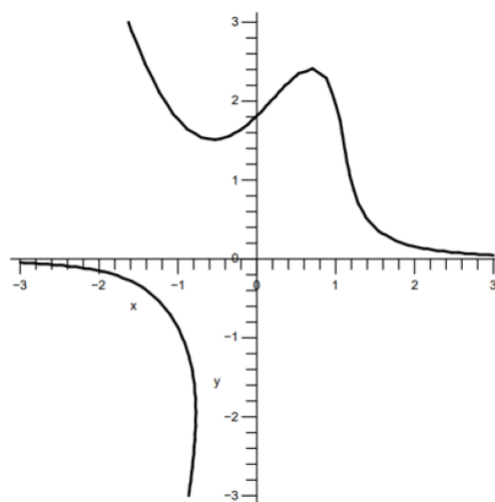


Workshop 9

Problem 1

Keywords: Implicit Differentiation, Linear Approximation

Consider the curve $5x^3y - 3xy^2 + y^3 = 6$, pictured below. The point $(1, 2)$ is on this curve.



- (a) Find an equation of the tangent line to the curve at the point $(1, 2)$.
- (b) There's a nearby point on the curve whose coordinates are $(1.07, u)$. Using linear approximation, find an approximate value for u .
- (c) There's a nearby point on the curve whose coordinates are $(v, 2.04)$. Using linear approximation, find an approximate value for v .

Problem 2

Keywords: Linear Approximation, Second Derivative

Consider the functions $f(x) = x^2$ and $g(x) = \sqrt{x}$.

- (a) Using linear approximation, find an approximate value of $f(3.01)$. Use a calculator to say whether the approximate value is larger or smaller than the actual value.
- (b) Using linear approximation, find an approximate value of $g(4.01)$. Use a calculator to say whether the approximate value is larger or smaller than the actual value.
- (c) We now try to explain what is happening. Compute the values of $f''(3)$ and $g''(4)$. Are they positive or negative?

- (d) Using the fact that the second derivative is the rate of change of the first derivative, explain why your approximated values were larger or smaller than the actual values for the functions f and g .

Problem 3

Keywords: Extreme Values

Let $f(x) = x^a(1 - x)^b$ where a and b are any two positive numbers. Use f in parts (a)-(d).

- (a) If $a = 3$ and $b = 6$, find the absolute maximum value and the absolute minimum value of f on $[0, 1]$.
- (b) If a and b are any positive numbers, find the critical points of f in the interval $(0, 1)$
- (c) If a and b are any positive numbers, find the absolute maximum and the absolute minimum value of f on $[0, 1]$.

Optional Problem (*just for fun, no need to submit*)

Consider a function f such that $f(1) = 6$. Alice had access to an explicit formula for the function. She used linear approximation to estimate the value of $f(0.98)$ and reported the answer 6.06. Assuming Alice is correct, find $f'(1)$ and use linear approximation to estimate the value of $\frac{1}{f(1.03)}$.